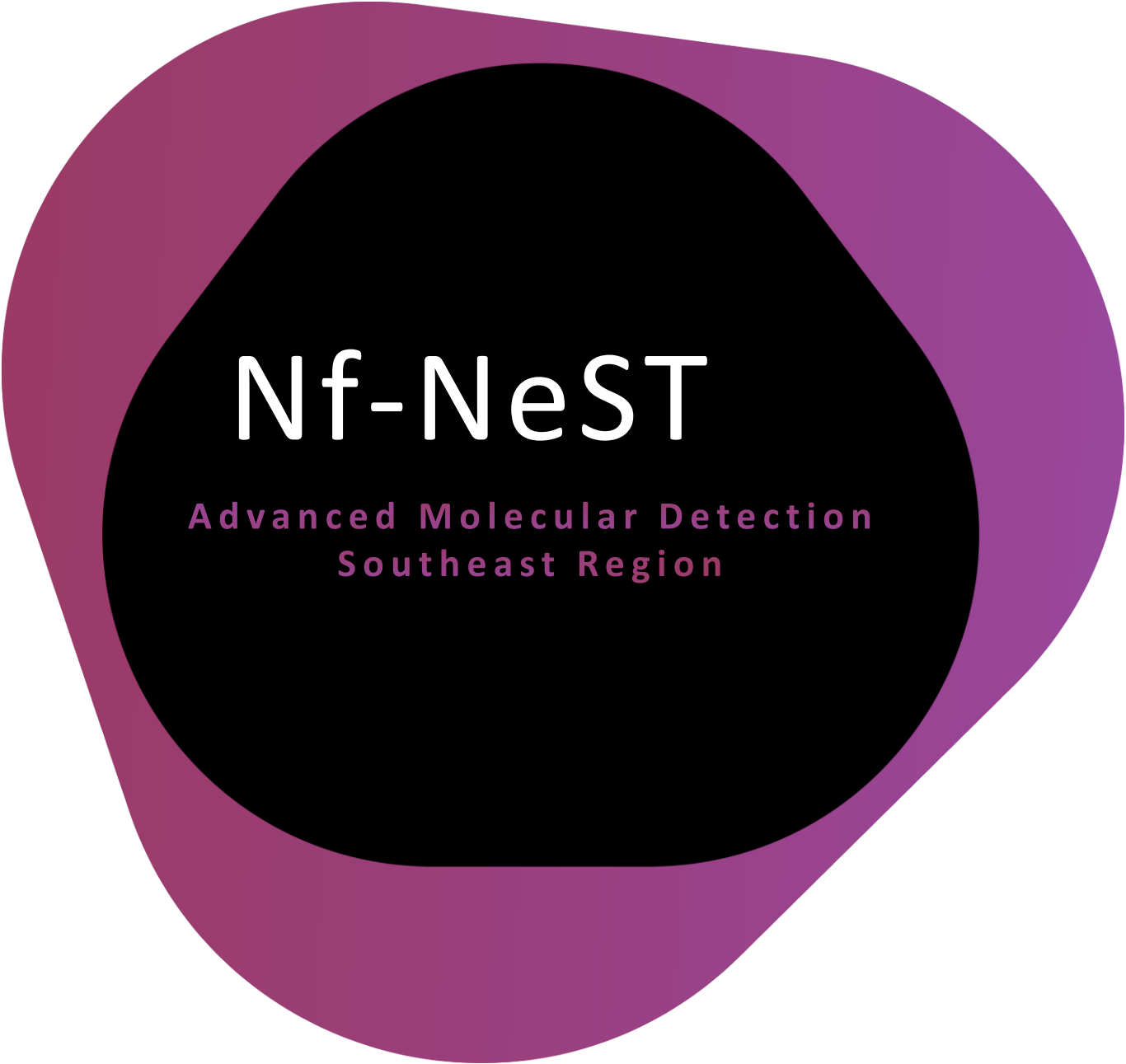


The logo consists of a central black circle with a white border. This circle is surrounded by a larger, irregular purple shape that has a white border. The text "Nf-NeST" is written in white, sans-serif font inside the black circle.

Nf-NeST

Advanced Molecular Detection
Southeast Region

This resource was made possible through funding provided under the Epidemiology and Laboratory Capacity for Prevention and Control of Emerging Infectious Diseases (ELC) Cooperative Agreement (CK24-0002), Project D: Advanced Molecular Detection to the Florida Department of Health. The conclusions, findings, and opinions expressed by authors do not necessarily reflect the official position of the U.S. Department of Health and Human Services, the Public Health Service, or the Centers for Disease Control and Prevention.

UPDATES

- BP3 Project D. AMD Updates – budgets, milestones, project approaches due in CAMP in **21 days!!**
- Internally your deadlines may be different
 - Check in with your ELC officer!
- Apply for BTL In-person Training travel for 2 Bioinformaticians in 1b.
 - This is a 2-day workshop in Athens, GA
- Moving to once-a-month meeting occurrence!

OVERVIEW

Purpose:

- This Nextflow pipeline processes Illumina sequencing reads for *Plasmodium falciparum* (Malaria) targeted sequencing data to generate standardized outputs, including trimmed reads, alignments, variant calls, annotations, and summary reports.

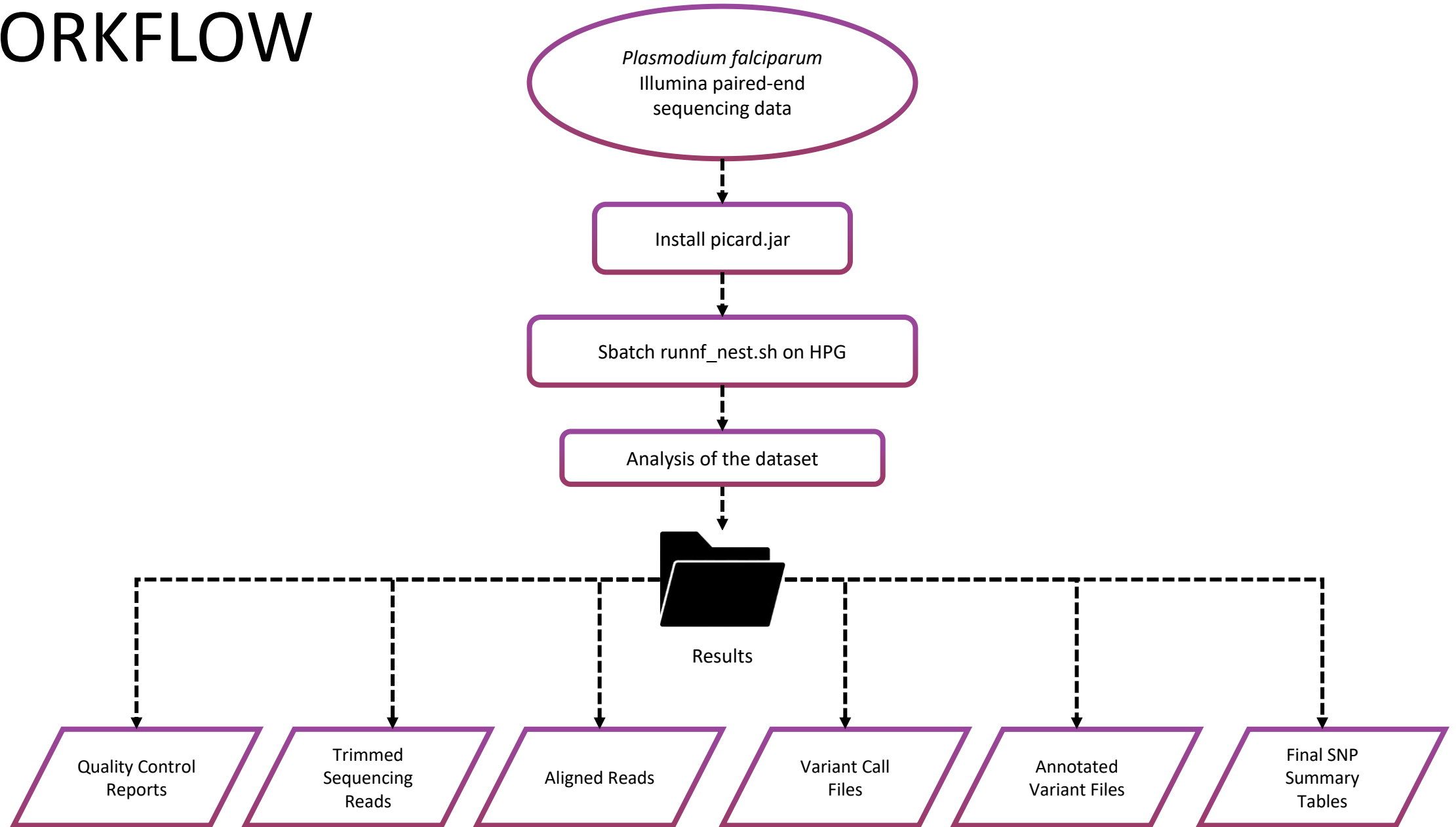
Usage:

- It enables public health laboratories and researchers to identify and analyze genetic variants associated with antimalarial drug resistance for genomic surveillance, variant interpretation, and downstream epidemiological analysis.

Dependencies:

- Nextflow
- Java
- Apptainer/Singularity
- Container containing BBTools, Bowtie2, Samtools, BCFtools, FreeBayes, FastQC, and GATK

WORKFLOW



Application

OBJECTIVE

Use a Malaria dataset and analyze the dataset using Nf-NeST pipeline

Application Cont.

```
cd /blue/bphl-<state>/<user>/repos/bphl-molecular/  
git clone https://github.com/CDCgov/ Nf-NeST  
mkdir analysis/  
cd analysis/  
cp /blue/bphl-<state>/<user>/repos/bphl-molecular/Nf-  
NeST/*
```

/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/Nf-NeST0312/

Name	Size	Changed	Rights	Owner
..		3/12/2026 9:26:09 PM	rw-r--r--	n.yengal...
6genes_ver3		3/12/2026 9:24:14 PM	rw-r--r--	n.yengal...
angola		3/12/2026 9:24:14 PM	rw-r--r--	n.yengal...
archive		3/12/2026 9:24:44 PM	rw-r--r--	n.yengal...
data		3/12/2026 9:24:45 PM	rw-r--r--	n.yengal...
inputfiles		3/13/2026 11:56:49 AM	rw-r--r--	n.yengal...
misc-fastq		3/12/2026 9:47:37 PM	rw-r--r--	n.yengal...
new6genes		3/12/2026 9:24:45 PM	rw-r--r--	n.yengal...
post_analysis		3/12/2026 10:14:57 PM	rw-r--r--	n.yengal...
pyscripts		3/13/2026 11:59:50 AM	rw-r--r--	n.yengal...
ref		3/13/2026 1:09:26 AM	rw-r--r--	n.yengal...
results		3/13/2026 12:00:11 PM	rw-r--r--	n.yengal...
singularity_cache		3/13/2026 11:56:32 AM	rw-r--r--	n.yengal...
testrun		3/13/2026 12:06:46 AM	rw-r--r--	n.yengal...
tmp		3/13/2026 12:00:11 PM	rw-r--r--	n.yengal...
work		3/13/2026 11:59:57 AM	rw-r--r--	n.yengal...
Dockerfile	3 KB	3/12/2026 9:24:14 PM	rw-r--r--	n.yengal...
nfneST.27175426.err	2 KB	3/13/2026 11:56:41 AM	rw-r--r--	n.yengal...
nfneST.27175426.out	200 KB	3/13/2026 12:00:11 PM	rw-r--r--	n.yengal...
nfNeST.nf	17 KB	3/12/2026 9:24:45 PM	rw-r--r--	n.yengal...
nfNeST_ver02.nf	15 KB	3/12/2026 9:24:45 PM	rw-r--r--	n.yengal...
nfNeST_ver03.nf	20 KB	3/13/2026 11:54:09 AM	rw-r--r--	n.yengal...
picard.jar	61,443 KB	11/14/2023 2:01:06 PM	rw-r--r--	n.yengal...
post_analysis_pool.py	11 KB	3/12/2026 9:24:45 PM	rw-r--r--	n.yengal...
PostAnalysisSOP.md	1 KB	3/12/2026 9:24:14 PM	rw-r--r--	n.yengal...
README.md	10 KB	3/12/2026 9:24:14 PM	rw-r--r--	n.yengal...
runnf_nest.sh	3 KB	3/13/2026 1:49:32 AM	rw-r--r--	n.yengal...
TODO.md	1 KB	3/12/2026 9:24:14 PM	rw-r--r--	n.yengal...
visualization_general.py	10 KB	3/12/2026 9:24:51 PM	rw-r--r--	n.yengal...

/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/Nf-NeST0312/inputfiles/

Name	Size	Changed	Rights	Owner
..		3/13/2026 11:56:43 AM	rw-r--r--	n.yengal...
output		3/13/2026 12:00:01 PM	rw-r--r--	n.yengal...
nextflow.config	1 KB	3/12/2026 11:20:24 PM	rw-r--r--	n.yengal...
SRR37504816_R1_001...	30,321 KB	3/12/2026 9:45:32 PM	rw-r--r--	n.yengal...
SRR37504816_R2_001...	32,078 KB	3/12/2026 9:45:32 PM	rw-r--r--	n.yengal...
SRR37504817_R1_001...	25,479 KB	3/12/2026 9:45:39 PM	rw-r--r--	n.yengal...
SRR37504817_R2_001...	26,839 KB	3/12/2026 9:45:39 PM	rw-r--r--	n.yengal...
SRR37504818_R1_001...	8,899 KB	3/12/2026 9:45:45 PM	rw-r--r--	n.yengal...
SRR37504818_R2_001...	9,522 KB	3/12/2026 9:45:45 PM	rw-r--r--	n.yengal...
SRR37504819_R1_001...	25,986 KB	3/12/2026 9:45:51 PM	rw-r--r--	n.yengal...
SRR37504819_R2_001...	27,505 KB	3/12/2026 9:45:51 PM	rw-r--r--	n.yengal...
SRR37504820_R1_001...	22,673 KB	3/12/2026 9:45:58 PM	rw-r--r--	n.yengal...
SRR37504820_R2_001...	24,079 KB	3/12/2026 9:45:58 PM	rw-r--r--	n.yengal...
SRR37504821_R1_001...	29,863 KB	3/12/2026 9:46:04 PM	rw-r--r--	n.yengal...
SRR37504821_R2_001...	31,584 KB	3/12/2026 9:46:04 PM	rw-r--r--	n.yengal...
SRR37504822_R1_001...	16,312 KB	3/12/2026 9:46:12 PM	rw-r--r--	n.yengal...
SRR37504822_R2_001...	16,594 KB	3/12/2026 9:46:12 PM	rw-r--r--	n.yengal...
SRR37504823_R1_001...	36,317 KB	3/12/2026 9:46:18 PM	rw-r--r--	n.yengal...
SRR37504823_R2_001...	38,578 KB	3/12/2026 9:46:18 PM	rw-r--r--	n.yengal...
SRR37504824_R1_001...	34,155 KB	3/12/2026 9:46:26 PM	rw-r--r--	n.yengal...
SRR37504824_R2_001...	36,230 KB	3/12/2026 9:46:26 PM	rw-r--r--	n.yengal...
SRR37504825_R1_001...	30,583 KB	3/12/2026 9:46:35 PM	rw-r--r--	n.yengal...
SRR37504825_R2_001...	32,702 KB	3/12/2026 9:46:35 PM	rw-r--r--	n.yengal...

Application Cont.

wget <https://github.com/broadinstitute/picard/releases/download/3.1.1/picard.jar>

↓

/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/Nf-NeST0312/				
Name	Size	Changed	Rights	Owner
..		3/12/2026 9:26:09 PM	rw-r--r--	n.yengal...
6genes_ver3		3/12/2026 9:24:14 PM	rw-r--r--	n.yengal...
angola		3/12/2026 9:24:14 PM	rw-r--r--	n.yengal...
archive		3/12/2026 9:24:44 PM	rw-r--r--	n.yengal...
data		3/12/2026 9:24:45 PM	rw-r--r--	n.yengal...
inputfiles		3/13/2026 11:56:49 AM	rw-r--r--	n.yengal...
misc-fastq		3/12/2026 9:47:37 PM	rw-r--r--	n.yengal...
new6genes		3/12/2026 9:24:45 PM	rw-r--r--	n.yengal...
post_analysis		3/12/2026 10:14:57 PM	rw-r--r--	n.yengal...
pyscripts		3/13/2026 11:59:50 AM	rw-r--r--	n.yengal...
ref		3/13/2026 1:09:26 AM	rw-r--r--	n.yengal...
results		3/13/2026 12:00:11 PM	rw-r--r--	n.yengal...
singularity_cache		3/13/2026 11:56:32 AM	rw-r--r--	n.yengal...
testrun		3/13/2026 12:06:46 AM	rw-r--r--	n.yengal...
tmp		3/13/2026 12:00:11 PM	rw-r--r--	n.yengal...
work		3/13/2026 11:59:57 AM	rw-r--r--	n.yengal...
Dockerfile	3 KB	3/12/2026 9:24:14 PM	rw-r--r--	n.yengal...
nfneSt.27175426.err	2 KB	3/13/2026 11:56:41 AM	rw-r--r--	n.yengal...
nfneSt.27175426.out	200 KB	3/13/2026 12:00:11 PM	rw-r--r--	n.yengal...
nfNeST.nf	17 KB	3/12/2026 9:24:45 PM	rw-r--r--	n.yengal...
nfNeST_ver02.nf	15 KB	3/12/2026 9:24:45 PM	rw-r--r--	n.yengal...
nfNeST_ver03.nf	20 KB	3/13/2026 11:54:09 AM	rw-r--r--	n.yengal...
picard.jar	61,443 KB	11/14/2023 2:01:06 PM	rw-r--r--	n.yengal...
post_analysis_pool.py	11 KB	3/12/2026 9:24:45 PM	rw-r--r--	n.yengal...
PostAnalysisSOP.md	1 KB	3/12/2026 9:24:14 PM	rw-r--r--	n.yengal...
README.md	10 KB	3/12/2026 9:24:14 PM	rw-r--r--	n.yengal...
runnf_nest.sh	3 KB	3/13/2026 1:49:32 AM	rw-r--r--	n.yengal...
TODO.md	1 KB	3/12/2026 9:24:14 PM	rw-r--r--	n.yengal...
visualization_general.py	10 KB	3/12/2026 9:24:51 PM	rw-r--r--	n.yengal...

Application Cont.

/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/Nf-NeST0312/

Name	Size	Changed	Rights	Owner
..		3/12/2026 9:26:09 PM	rw-r--r--	n.yengal...
6genes_ver3		3/12/2026 9:24:14 PM	rw-r--r--	n.yengal...
angola		3/12/2026 9:24:14 PM	rw-r--r--	n.yengal...
archive		3/12/2026 9:24:44 PM	rw-r--r--	n.yengal...
data		3/12/2026 9:24:45 PM	rw-r--r--	n.yengal...
inputfiles		3/13/2026 11:56:49 AM	rw-r--r--	n.yengal...
misc-fastq		3/12/2026 9:47:37 PM	rw-r--r--	n.yengal...
new6genes		3/12/2026 9:24:45 PM	rw-r--r--	n.yengal...
post_analysis		3/12/2026 10:14:57 PM	rw-r--r--	n.yengal...
pyscripts		3/13/2026 11:59:50 AM	rw-r--r--	n.yengal...
ref		3/13/2026 1:09:26 AM	rw-r--r--	n.yengal...
results		3/13/2026 12:00:11 PM	rw-r--r--	n.yengal...
singularity_cache		3/13/2026 11:56:32 AM	rw-r--r--	n.yengal...
testrun		3/13/2026 12:06:46 AM	rw-r--r--	n.yengal...
tmp		3/13/2026 12:00:11 PM	rw-r--r--	n.yengal...
work		3/13/2026 11:59:57 AM	rw-r--r--	n.yengal...
Dockerfile	3 KB	3/12/2026 9:24:14 PM	rw-r--r--	n.yengal...
nfneST.27175426.err	2 KB	3/13/2026 11:56:41 AM	rw-r--r--	n.yengal...
nfneST.27175426.out	200 KB	3/13/2026 12:00:11 PM	rw-r--r--	n.yengal...
nfNeST.nf	17 KB	3/12/2026 9:24:45 PM	rw-r--r--	n.yengal...
nfNeST_ver02.nf	15 KB	3/12/2026 9:24:45 PM	rw-r--r--	n.yengal...
nfNeST_ver03.nf	20 KB	3/13/2026 11:54:09 AM	rw-r--r--	n.yengal...
picard.jar	61,443 KB	11/14/2023 2:01:06 PM	rw-r--r--	n.yengal...
post_analysis_pool.py	11 KB	3/12/2026 9:24:45 PM	rw-r--r--	n.yengal...
PostAnalysisSOP.md	1 KB	3/12/2026 9:24:14 PM	rw-r--r--	n.yengal...
README.md	10 KB	3/12/2026 9:24:14 PM	rw-r--r--	n.yengal...
runnf_nest.sh	3 KB	3/13/2026 1:49:32 AM	rw-r--r--	n.yengal...
TODO.md	1 KB	3/12/2026 9:24:14 PM	rw-r--r--	n.yengal...
visualization_general.py	10 KB	3/12/2026 9:24:51 PM	rw-r--r--	n.yengal...

```
module purge
source /etc/profile.d/modules.sh

unset JAVA_HOME
unset JAVA_CMD

#Load modules
|
module load apptainer
module load nextflow/22.10.6

module load bbmap
module load bowtie2
module load samtools
module load bcftools
module load freebayes
module load fastqc
module load gatk
module load snpeff

# Load Java
module load java/17

export _JAVA_OPTIONS="-XX:UseAVX=0"

#Pipeline directory
PIPELINE_DIR=/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/Nf-NeST0312

cd $PIPELINE_DIR || exit 1

#Create directories
mkdir -p work
mkdir -p results
mkdir -p tmp
mkdir -p singularity_cache
mkdir -p .nextflow

#Nextflow environment
export NXF_HOME=$PIPELINE_DIR/.nextflow
export NXF_WORK=$PIPELINE_DIR/work
export NXF_TEMP=$PIPELINE_DIR/tmp

export NXF_SINGULARITY_CACHEDIR=$PIPELINE_DIR/singularity_cache
export NXF_SINGULARITY_AUTO_MOUNTS=true

# Required because pipeline uses DSL1
export NXF_DEFAULT_DSL=1

#Run pipeline

nextflow run nfNeST_ver03.nf \
  -c inputfiles/nextflow.config \
  -work-dir $PIPELINE_DIR/work \
  -with-report results/report.html \
  -with-trace results/trace.txt \
  -with-timeline results/timeline.html \
  -with-dag results/dag.html \
  -resume

EXITCODE=$?

exit $EXITCODE
```

Application Cont.

/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/Nf-NeST0312/				
Name	Size	Changed	Rights	Owner
..		3/12/2026 9:26:09 PM	rwxr-sr-x	n.yengal...
6genes_ver3		3/12/2026 9:24:14 PM	rwxr-sr-x	n.yengal...
angola		3/12/2026 9:24:14 PM	rwxr-sr-x	n.yengal...
archive		3/12/2026 9:24:44 PM	rwxr-sr-x	n.yengal...
data		3/12/2026 9:24:45 PM	rwxr-sr-x	n.yengal...
inputfiles		3/13/2026 11:56:49 AM	rwxr-sr-x	n.yengal...
misc-fastq		3/12/2026 9:47:37 PM	rwxr-sr-x	n.yengal...
new6genes		3/12/2026 9:24:45 PM	rwxr-sr-x	n.yengal...
post_analysis		3/12/2026 10:14:57 PM	rwxr-sr-x	n.yengal...
pyscripts		3/13/2026 11:59:50 AM	rwxr-sr-x	n.yengal...
ref		3/13/2026 1:09:26 AM	rwxr-sr-x	n.yengal...
results		3/13/2026 12:00:11 PM	rwxr-sr-x	n.yengal...
singularity_cache		3/13/2026 11:56:32 AM	rwxr-sr-x	n.yengal...
testrun		3/13/2026 12:06:46 AM	rwxr-sr-x	n.yengal...
tmp		3/13/2026 12:00:11 PM	rwxr-sr-x	n.yengal...
work		3/13/2026 11:59:57 AM	rwxr-sr-x	n.yengal...
Dockerfile	3 KB	3/12/2026 9:24:14 PM	rw-r--r--	n.yengal...
nfneSt.27175426.err	2 KB	3/13/2026 11:56:41 AM	rw-r--r--	n.yengal...
nfneSt.27175426.out	200 KB	3/13/2026 12:00:11 PM	rw-r--r--	n.yengal...
nfNeST.nf	17 KB	3/12/2026 9:24:45 PM	rw-r--r--	n.yengal...
nfNeST_ver02.nf	15 KB	3/12/2026 9:24:45 PM	rw-r--r--	n.yengal...
nfNeST_ver03.nf	20 KB	3/13/2026 11:54:09 AM	rw-r--r--	n.yengal...
picard.jar	61,443 KB	11/14/2023 2:01:06 PM	rw-r--r--	n.yengal...
post_analysis_pool.py	11 KB	3/12/2026 9:24:45 PM	rw-r--r--	n.yengal...
PostAnalysisSOP.md	1 KB	3/12/2026 9:24:14 PM	rw-r--r--	n.yengal...
README.md	10 KB	3/12/2026 9:24:14 PM	rw-r--r--	n.yengal...
runnf_nest.sh	3 KB	3/13/2026 1:49:32 AM	rw-r--r--	n.yengal...
TODO.md	1 KB	3/12/2026 9:24:14 PM	rw-r--r--	n.yengal...
visualization_general.py	10 KB	3/12/2026 9:24:51 PM	rw-r--r--	n.yengal...

/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/Nf-NeST0312/inputfiles/				
Name	Size	Changed	Rights	Owner
..		3/13/2026 11:56:43 AM	rwxr-sr-x	n.yengal...
output		3/13/2026 12:00:01 PM	rwxr-sr-x	n.yengal...
nextflow.config	1 KB	3/12/2026 11:20:24 PM	rw-r--r--	n.yengal...
SRR37504816_R1_001...	30,321 KB	3/12/2026 9:45:32 PM	rw-r--r--	n.yengal...
SRR37504816_R2_001...	32,078 KB	3/12/2026 9:45:32 PM	rw-r--r--	n.yengal...
SRR37504817_R1_001...	25,479 KB	3/12/2026 9:45:39 PM	rw-r--r--	n.yengal...
SRR37504817_R2_001...	26,839 KB	3/12/2026 9:45:39 PM	rw-r--r--	n.yengal...
SRR37504818_R1_001...	8,899 KB	3/12/2026 9:45:45 PM	rw-r--r--	n.yengal...
SRR37504818_R2_001...	9,522 KB	3/12/2026 9:45:45 PM	rw-r--r--	n.yengal...
SRR37504819_R1_001...	25,986 KB	3/12/2026 9:45:51 PM	rw-r--r--	n.yengal...
SRR37504819_R2_001...	27,505 KB	3/12/2026 9:45:51 PM	rw-r--r--	n.yengal...
SRR37504820_R1_001...	22,673 KB	3/12/2026 9:45:58 PM	rw-r--r--	n.yengal...
SRR37504820_R2_001...	24,079 KB	3/12/2026 9:45:58 PM	rw-r--r--	n.yengal...
SRR37504821_R1_001...	29,863 KB	3/12/2026 9:46:04 PM	rw-r--r--	n.yengal...
SRR37504821_R2_001...	31,584 KB	3/12/2026 9:46:04 PM	rw-r--r--	n.yengal...
SRR37504822_R1_001...	16,312 KB	3/12/2026 9:46:12 PM	rw-r--r--	n.yengal...
SRR37504822_R2_001...	16,594 KB	3/12/2026 9:46:12 PM	rw-r--r--	n.yengal...
SRR37504823_R1_001...	36,317 KB	3/12/2026 9:46:18 PM	rw-r--r--	n.yengal...
SRR37504823_R2_001...	38,578 KB	3/12/2026 9:46:18 PM	rw-r--r--	n.yengal...
SRR37504824_R1_001...	34,155 KB	3/12/2026 9:46:26 PM	rw-r--r--	n.yengal...
SRR37504824_R2_001...	36,230 KB	3/12/2026 9:46:26 PM	rw-r--r--	n.yengal...
SRR37504825_R1_001...	30,583 KB	3/12/2026 9:46:35 PM	rw-r--r--	n.yengal...
SRR37504825_R2_001...	32,702 KB	3/12/2026 9:46:35 PM	rw-r--r--	n.yengal...

/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/Nf-NeST0312/inputfiles/output/				
Name	Size	Changed	Rights	Owner
..		3/13/2026 11:56:49 AM	rwxr-sr-x	n.yengal...
alignedReads		3/13/2026 11:58:19 AM	rwxr-sr-x	n.yengal...
annotate		3/13/2026 11:59:22 AM	rwxr-sr-x	n.yengal...
Bowtie2Index		3/13/2026 11:56:51 AM	rwxr-sr-x	n.yengal...
combineFastq		3/13/2026 11:56:54 AM	rwxr-sr-x	n.yengal...
extract		3/13/2026 11:59:48 AM	rwxr-sr-x	n.yengal...
filter		3/13/2026 11:59:41 AM	rwxr-sr-x	n.yengal...
final_vcf		3/13/2026 11:59:35 AM	rwxr-sr-x	n.yengal...
forjson		3/13/2026 12:00:10 PM	rwxr-sr-x	n.yengal...
GenerateVCFSam		3/13/2026 11:59:05 AM	rwxr-sr-x	n.yengal...
log		3/13/2026 12:00:01 PM	rwxr-sr-x	n.yengal...
postFastqQC		3/13/2026 11:58:20 AM	rwxr-sr-x	n.yengal...
preFastqQC		3/13/2026 11:58:31 AM	rwxr-sr-x	n.yengal...
samtoolcoverage		3/13/2026 11:58:58 AM	rwxr-sr-x	n.yengal...
snpfilter		3/13/2026 12:00:10 PM	rwxr-sr-x	n.yengal...
spread		3/13/2026 11:59:57 AM	rwxr-sr-x	n.yengal...
trimFastq		3/13/2026 11:57:47 AM	rwxr-sr-x	n.yengal...
vartype		3/13/2026 11:59:27 AM	rwxr-sr-x	n.yengal...

sbatch runnf_nest.sh

Application Cont.

Time	Read	Base	SameReac	SameReac	Unfiltered	Unfiltered	Paired	Unpaired	AlConUni	AlConRep	AlConFail	AlDis	AlConFail	AlConFail	AlConFail	AlConRep	AlConRep	AlConRep	AlUnpUni	AlUnpRep	AlUnpFail	SeedSearch	NRange	NElt	IntraScacI	InterScacI	OutOfMer	AlBWOp
1.8E+09	56188	1.7E+07	0	0	112376	1.7E+07	56188	0	3944	0	52244	0	11	0	104477	0	0	0	0	0	0	3716890	128426	134000	0	0	0	2172768
1.8E+09	87152	2.6E+07	0	0	174304	2.6E+07	87152	0	6037	0	81115	0	24	0	162206	0	0	0	0	0	0	5764904	196560	205772	0	0	0	3335375
1.8E+09	156869	4.7E+07	0	0	313738	4.7E+07	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1E+07	355148	371470	0	0	0	6022913

Name	Size	Changed	Rights	Owner
..		3/13/2026 11:56:49 AM	rw-r--r--	n.yengal...
alignedReads		3/13/2026 11:58:19 AM	rw-r--r--	n.yengal...
annotate		3/13/2026 11:59:22 AM	rw-r--r--	n.yengal...
Bowtie2Index		3/13/2026 11:56:51 AM	rw-r--r--	n.yengal...
combineFastq		3/13/2026 11:56:54 AM	rw-r--r--	n.yengal...
extract		3/13/2026 11:59:48 AM	rw-r--r--	n.yengal...
filter		3/13/2026 11:59:41 AM	rw-r--r--	n.yengal...
final_vcf		3/13/2026 11:59:35 AM	rw-r--r--	n.yengal...
forjson		3/13/2026 12:00:10 PM	rw-r--r--	n.yengal...
GenerateVCFSam		3/13/2026 11:59:05 AM	rw-r--r--	n.yengal...
log		3/13/2026 12:00:01 PM	rw-r--r--	n.yengal...
postFastqQC		3/13/2026 11:58:20 AM	rw-r--r--	n.yengal...
preFastqQC		3/13/2026 11:58:31 AM	rw-r--r--	n.yengal...
samtoolcoverage		3/13/2026 11:58:58 AM	rw-r--r--	n.yengal...
snpfilter		3/13/2026 12:00:10 PM	rw-r--r--	n.yengal...
spread		3/13/2026 11:59:57 AM	rw-r--r--	n.yengal...
trimFastq		3/13/2026 11:57:47 AM	rw-r--r--	n.yengal...
vartype		3/13/2026 11:59:27 AM	rw-r--r--	n.yengal...

Name	Size	Changed	Rights	Owner
..		3/13/2026 11:58:19 AM	rw-r--r--	n.yengal...
Stats		3/13/2026 11:58:10 AM	rw-r--r--	n.yengal...
SRR37504816.sam	108,346 KB	3/13/2026 11:58:10 AM	rw-r--r--	n.yengal...
SRR37504816_SR.bam	12,638 KB	3/13/2026 11:58:29 AM	rw-r--r--	n.yengal...

Name	Size	Changed	Rights	Owner
..		3/13/2026 11:58:29 AM	rw-r--r--	n.yengal...
SRR37504816_bmet.txt	3 KB	3/13/2026 11:58:10 AM	rw-r--r--	n.yengal...

Application Cont.

/.../n.yengalareddy/repos/bphl-molecular/analysis/Nf-NeST0312/inputfiles/output/Bowtie2Index/

Name	Size	Changed	Rights	Owner
..		3/13/2026 12:00:01 PM	rw-r--r--	n.yengal...
genome.index.1.bt2	4,103 KB	3/13/2026 11:56:51 AM	rw-r--r--	n.yengal...
genome.index.2.bt2	5 KB	3/13/2026 11:56:50 AM	rw-r--r--	n.yengal...
genome.index.3.bt2	1 KB	3/13/2026 11:56:50 AM	rw-r--r--	n.yengal...
genome.index.4.bt2	5 KB	3/13/2026 11:56:51 AM	rw-r--r--	n.yengal...
genome.index.rev.1.bt2	4,103 KB	3/13/2026 11:56:51 AM	rw-r--r--	n.yengal...
genome.index.rev.2.bt2	5 KB	3/13/2026 11:56:50 AM	rw-r--r--	n.yengal...

/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/Nf-NeST0312/inputfiles/output/

Name	Size	Changed	Rights	Owner
..		3/13/2026 11:56:49 AM	rw-r--r--	n.yengal...
alignedReads		3/13/2026 11:58:19 AM	rw-r--r--	n.yengal...
annotate		3/13/2026 11:59:22 AM	rw-r--r--	n.yengal...
Bowtie2Index		3/13/2026 11:56:51 AM	rw-r--r--	n.yengal...
combineFastq		3/13/2026 11:56:54 AM	rw-r--r--	n.yengal...
extract		3/13/2026 11:59:48 AM	rw-r--r--	n.yengal...
filter		3/13/2026 11:59:41 AM	rw-r--r--	n.yengal...
final_vcf		3/13/2026 11:59:35 AM	rw-r--r--	n.yengal...
forjson		3/13/2026 12:00:10 PM	rw-r--r--	n.yengal...
GenerateVCFSam		3/13/2026 11:59:05 AM	rw-r--r--	n.yengal...
log		3/13/2026 12:00:01 PM	rw-r--r--	n.yengal...
postFastqQC		3/13/2026 11:58:20 AM	rw-r--r--	n.yengal...
preFastqQC		3/13/2026 11:58:31 AM	rw-r--r--	n.yengal...
samtoolcoverage		3/13/2026 11:58:58 AM	rw-r--r--	n.yengal...
snpfilter		3/13/2026 12:00:10 PM	rw-r--r--	n.yengal...
spread		3/13/2026 11:59:57 AM	rw-r--r--	n.yengal...
trimFastq		3/13/2026 11:57:47 AM	rw-r--r--	n.yengal...
vartype		3/13/2026 11:59:27 AM	rw-r--r--	n.yengal...



Application Cont.

/.../repos/bphl-molecular/analysis/Nf-NeST0312/inputfiles/output/combineFastq/SRR37504816/

Name	Size	Changed	Rights	Owner
..		3/13/2026 11:56:54 AM	rw-r--r--	n.yengal...
SRR37504816_R1.fast...	30,321 KB	3/13/2026 11:56:51 AM	rw-r--r--	n.yengal...
SRR37504816_R2.fast...	32,078 KB	3/13/2026 11:56:51 AM	rw-r--r--	n.yengal...

/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/Nf-NeST0312/inputfiles/output/

Name	Size	Changed	Rights	Owner
..		3/13/2026 11:56:49 AM	rw-r--r--	n.yengal...
alignedReads		3/13/2026 11:58:19 AM	rw-r--r--	n.yengal...
annotate		3/13/2026 11:59:22 AM	rw-r--r--	n.yengal...
Bowtie2Index		3/13/2026 11:56:51 AM	rw-r--r--	n.yengal...
combineFastq		3/13/2026 11:56:54 AM	rw-r--r--	n.yengal...
extract		3/13/2026 11:59:48 AM	rw-r--r--	n.yengal...
filter		3/13/2026 11:59:41 AM	rw-r--r--	n.yengal...
final_vcf		3/13/2026 11:59:35 AM	rw-r--r--	n.yengal...
forjson		3/13/2026 12:00:10 PM	rw-r--r--	n.yengal...
GenerateVCFSam		3/13/2026 11:59:05 AM	rw-r--r--	n.yengal...
log		3/13/2026 12:00:01 PM	rw-r--r--	n.yengal...
postFastqQC		3/13/2026 11:58:20 AM	rw-r--r--	n.yengal...
preFastqQC		3/13/2026 11:58:31 AM	rw-r--r--	n.yengal...
samtoolcoverage		3/13/2026 11:58:58 AM	rw-r--r--	n.yengal...
snfilter		3/13/2026 12:00:10 PM	rw-r--r--	n.yengal...
spread		3/13/2026 11:59:57 AM	rw-r--r--	n.yengal...
trimFastq		3/13/2026 11:57:47 AM	rw-r--r--	n.yengal...
vartype		3/13/2026 11:59:27 AM	rw-r--r--	n.yengal...



Application Cont.

/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/Nf-NeST0312/inputfiles/output/

Name	Size	Changed	Rights	Owner
..		3/13/2026 11:56:49 AM	rwxr-sr-x	n.yengal...
alignedReads		3/13/2026 11:58:19 AM	rwxr-sr-x	n.yengal...
annotate		3/13/2026 11:59:22 AM	rwxr-sr-x	n.yengal...
Bowtie2Index		3/13/2026 11:56:51 AM	rwxr-sr-x	n.yengal...
combineFastq		3/13/2026 11:56:54 AM	rwxr-sr-x	n.yengal...
extract		3/13/2026 11:59:48 AM	rwxr-sr-x	n.yengal...
filter		3/13/2026 11:59:41 AM	rwxr-sr-x	n.yengal...
final_vcf		3/13/2026 11:59:35 AM	rwxr-sr-x	n.yengal...
forjson		3/13/2026 12:00:10 PM	rwxr-sr-x	n.yengal...
GenerateVCFSam		3/13/2026 11:59:05 AM	rwxr-sr-x	n.yengal...
log		3/13/2026 12:00:01 PM	rwxr-sr-x	n.yengal...
postFastqQC		3/13/2026 11:58:20 AM	rwxr-sr-x	n.yengal...
preFastqQC		3/13/2026 11:58:31 AM	rwxr-sr-x	n.yengal...
samtoolcoverage		3/13/2026 11:58:58 AM	rwxr-sr-x	n.yengal...
snfilter		3/13/2026 12:00:10 PM	rwxr-sr-x	n.yengal...
spread		3/13/2026 11:59:57 AM	rwxr-sr-x	n.yengal...
trimFastq		3/13/2026 11:57:47 AM	rwxr-sr-x	n.yengal...
vartype		3/13/2026 11:59:27 AM	rwxr-sr-x	n.yengal...

/.../repos/bphl-molecular/analysis/Nf-NeST0312/inputfiles/output/final_vcf/SRR37504816/

Name	Size	Changed	Rights	Owner
..		3/13/2026 11:59:35 AM	rwxr-sr-x	n.yengal...
final_SRR37504816.vcf	1 KB	3/13/2026 11:59:31 AM	rw-r--r--	n.yengal...



Application Cont.

/.../bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/Nf-NeST0312/inputfiles/output/forjson/

Name	Size	Changed	Rights	Owner
..		3/13/2026 12:00:01 PM	rw-r--r--	n.yengal...
SRR37504816.csv	5 KB	3/13/2026 12:00:02 PM	rw-r--r--	n.yengal...
SRR37504817.csv	5 KB	3/13/2026 12:00:01 PM	rw-r--r--	n.yengal...
SRR37504818.csv	5 KB	3/13/2026 12:00:01 PM	rw-r--r--	n.yengal...
SRR37504819.csv	5 KB	3/13/2026 12:00:06 PM	rw-r--r--	n.yengal...
SRR37504820.csv	5 KB	3/13/2026 12:00:10 PM	rw-r--r--	n.yengal...
SRR37504821.csv	5 KB	3/13/2026 12:00:02 PM	rw-r--r--	n.yengal...
SRR37504822.csv	5 KB	3/13/2026 12:00:01 PM	rw-r--r--	n.yengal...
SRR37504823.csv	5 KB	3/13/2026 12:00:08 PM	rw-r--r--	n.yengal...
SRR37504824.csv	5 KB	3/13/2026 12:00:01 PM	rw-r--r--	n.yengal...
SRR37504825.csv	5 KB	3/13/2026 12:00:03 PM	rw-r--r--	n.yengal...

/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/Nf-NeST0312/inputfiles/output/

Name	Size	Changed	Rights	Owner
..		3/13/2026 11:56:49 AM	rw-r--r--	n.yengal...
alignedReads		3/13/2026 11:58:19 AM	rw-r--r--	n.yengal...
annotate		3/13/2026 11:59:22 AM	rw-r--r--	n.yengal...
Bowtie2Index		3/13/2026 11:56:51 AM	rw-r--r--	n.yengal...
combineFastq		3/13/2026 11:56:54 AM	rw-r--r--	n.yengal...
extract		3/13/2026 11:59:48 AM	rw-r--r--	n.yengal...
filter		3/13/2026 11:59:41 AM	rw-r--r--	n.yengal...
final_vcf		3/13/2026 11:59:35 AM	rw-r--r--	n.yengal...
forjson		3/13/2026 12:00:10 PM	rw-r--r--	n.yengal...
GenerateVCFSam		3/13/2026 11:59:05 AM	rw-r--r--	n.yengal...
log		3/13/2026 12:00:01 PM	rw-r--r--	n.yengal...
postFastqQC		3/13/2026 11:58:20 AM	rw-r--r--	n.yengal...
preFastqQC		3/13/2026 11:58:31 AM	rw-r--r--	n.yengal...
samtoolcoverage		3/13/2026 11:58:58 AM	rw-r--r--	n.yengal...
snpfilter		3/13/2026 12:00:10 PM	rw-r--r--	n.yengal...
spread		3/13/2026 11:59:57 AM	rw-r--r--	n.yengal...
trimFastq		3/13/2026 11:57:47 AM	rw-r--r--	n.yengal...
vartype		3/13/2026 11:59:27 AM	rw-r--r--	n.yengal...



Advanced Molecular Detection
Southeast Region Bioinformatics

[/blue/bphl-florida/n.vengalareddy/repos/bphl-molecular/analysis/Nf-NeST0312/inputfiles/output/](#)

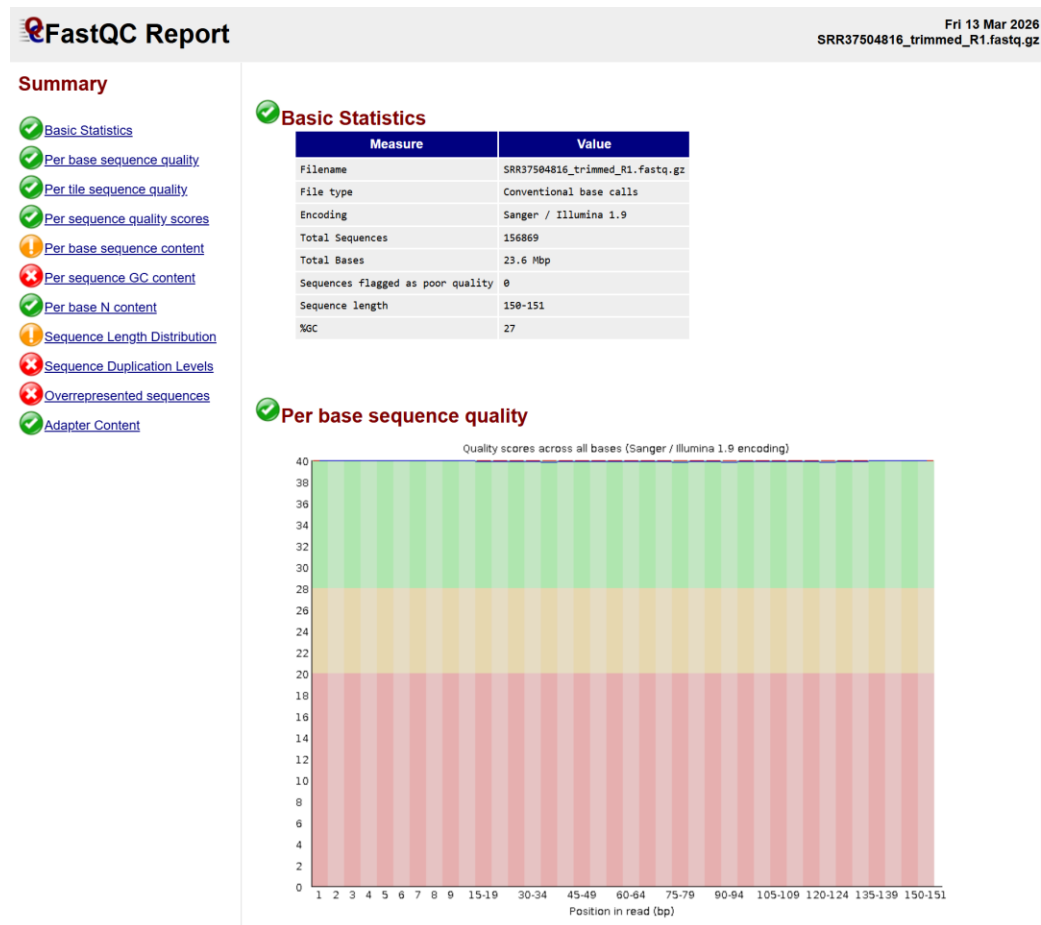
```

/.../repos/bphl-molecular/analysis/Nf-NeST0312/inputfiles/output/GenerateVCFSam/SRR37504816/

```

Name	Size	Changed	Rights	Owner
...		3/13/2026 11:59:05 AM	rxwxr-sr-x	n.yengal...
SRR37504816-1.vcf	5 KB	3/13/2026 11:59:00 AM	rw-r--r--	n.yengal...
SRR37504816-2.vcf	14 KB	3/13/2026 11:59:00 AM	rw-r--r--	n.yengal...
SRR37504816-3.vcf	0 KB	3/13/2026 11:59:00 AM	rw-r--r--	n.yengal...

Application Cont.



Name	Size	Changed	Rights	Owner
..		3/13/2026 11:56:49 AM	rw-r--r--	n.yengal...
alignedReads		3/13/2026 11:58:19 AM	rw-r--r--	n.yengal...
annotate		3/13/2026 11:59:22 AM	rw-r--r--	n.yengal...
Bowtie2Index		3/13/2026 11:56:51 AM	rw-r--r--	n.yengal...
combineFastq		3/13/2026 11:56:54 AM	rw-r--r--	n.yengal...
extract		3/13/2026 11:59:48 AM	rw-r--r--	n.yengal...
filter		3/13/2026 11:59:41 AM	rw-r--r--	n.yengal...
final_vcf		3/13/2026 11:59:35 AM	rw-r--r--	n.yengal...
forjson		3/13/2026 12:00:10 PM	rw-r--r--	n.yengal...
GenerateVCFSam		3/13/2026 11:59:05 AM	rw-r--r--	n.yengal...
log		3/13/2026 12:00:01 PM	rw-r--r--	n.yengal...
postFastqQC		3/13/2026 11:58:20 AM	rw-r--r--	n.yengal...
preFastqQC		3/13/2026 11:58:31 AM	rw-r--r--	n.yengal...
samtoolcoverage		3/13/2026 11:58:58 AM	rw-r--r--	n.yengal...
snpfilter		3/13/2026 12:00:10 PM	rw-r--r--	n.yengal...
spread		3/13/2026 11:59:57 AM	rw-r--r--	n.yengal...
trimFastq		3/13/2026 11:57:47 AM	rw-r--r--	n.yengal...
vartype		3/13/2026 11:59:27 AM	rw-r--r--	n.yengal...

Name	Size	Changed	Rights	Owner
..		3/13/2026 11:58:16 AM	rw-r--r--	n.yengal...
Icons		3/13/2026 11:58:16 AM	rw-r--r--	n.yengal...
Images		3/13/2026 11:58:16 AM	rw-r--r--	n.yengal...
fastqc.fo	67 KB	3/13/2026 11:58:16 AM	rw-r--r--	n.yengal...
fastqc_data.txt	145 KB	3/13/2026 11:58:16 AM	rw-r--r--	n.yengal...
fastqc_report.html	564 KB	3/13/2026 11:58:16 AM	rw-r--r--	n.yengal...
summary.txt	1 KB	3/13/2026 11:58:16 AM	rw-r--r--	n.yengal...

Application Cont.

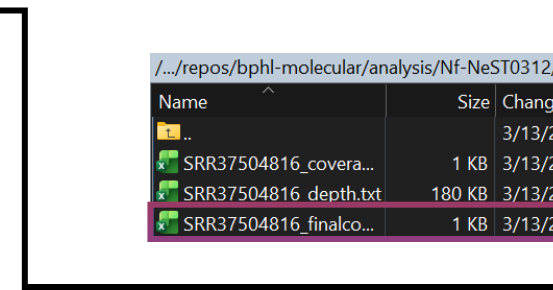
#rname	startpos	endpos	numreads	covbases	coverage	meandepth	meanbase	meanmapq
DHFR	1	1827	2279	704	38.5331	188.325	39.9	42
DHPS	1	2417	4099	586	24.2449	256.026	39.9	42
K13	1	2181	7252	1297	59.4681	501.979	39.9	42
PfCRT	1	3399	2528	852	25.0662	112.125	40	41
PfMDR1	1	4260	5697	1245	29.2254	201.881	39.9	42
mitochond	1	5967	0	0	0	0	0	0

/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/Nf-NeST0312/inputfiles/output/

Name	Size	Changed	Rights	Owner
..		3/13/2026 11:56:49 AM	rw-r--r--	n.yengal...
alignedReads		3/13/2026 11:58:19 AM	rw-r--r--	n.yengal...
annotate		3/13/2026 11:59:22 AM	rw-r--r--	n.yengal...
Bowtie2Index		3/13/2026 11:56:51 AM	rw-r--r--	n.yengal...
combineFastq		3/13/2026 11:56:54 AM	rw-r--r--	n.yengal...
extract		3/13/2026 11:59:48 AM	rw-r--r--	n.yengal...
filter		3/13/2026 11:59:41 AM	rw-r--r--	n.yengal...
final_vcf		3/13/2026 11:59:35 AM	rw-r--r--	n.yengal...
forjson		3/13/2026 12:00:10 PM	rw-r--r--	n.yengal...
GenerateVCFSam		3/13/2026 11:59:05 AM	rw-r--r--	n.yengal...
log		3/13/2026 12:00:01 PM	rw-r--r--	n.yengal...
postFastqQC		3/13/2026 11:58:20 AM	rw-r--r--	n.yengal...
preFastqQC		3/13/2026 11:58:31 AM	rw-r--r--	n.yengal...
samtoolcoverage		3/13/2026 11:58:58 AM	rw-r--r--	n.yengal...
snpfilter		3/13/2026 12:00:10 PM	rw-r--r--	n.yengal...
spread		3/13/2026 11:59:57 AM	rw-r--r--	n.yengal...
trimFastq		3/13/2026 11:57:47 AM	rw-r--r--	n.yengal...
vartype		3/13/2026 11:59:27 AM	rw-r--r--	n.yengal...

/.../repos/bphl-molecular/analysis/Nf-NeST0312/inputfiles/output/samtoolcoverage/SRR37504816/

Name	Size	Changed	Rights	Owner
..		3/13/2026 11:58:58 AM	rw-r--r--	n.yengal...
SRR37504816_cove...	1 KB	3/13/2026 11:58:50 AM	rw-r--r--	n.yengal...
SRR37504816_depth.txt	180 KB	3/13/2026 11:58:50 AM	rw-r--r--	n.yengal...
SRR37504816_finalco...	1 KB	3/13/2026 11:58:50 AM	rw-r--r--	n.yengal...



Conclusion



Fundamentals of Nf-NeST



Installation and setup of Nf-NeST in HPG



Successfully executed job query for Nf-NeST



Generated output files



Advanced Molecular Detection

Southeast Region Bioinformatics

Questions?

bphl-sebioinformatics@flhealth.gov

Molly Mitchell, PhD

Bioinformatics Supervisor

Molly.Mitchell@flhealth.gov

Nikhil Reddy, MS

Bioinformatician

Nikhil.Yengala@flhealth.gov

Sam Bernhoft, MPH

Bioinformatician

Samantha.bernhoft@flhealth.gov